

DATA WAREHOUSING AND DATA MINING

DAY - 1 LAB PROGRAMS

NAME: B KIRTHANAA

REG. NO – 192324244

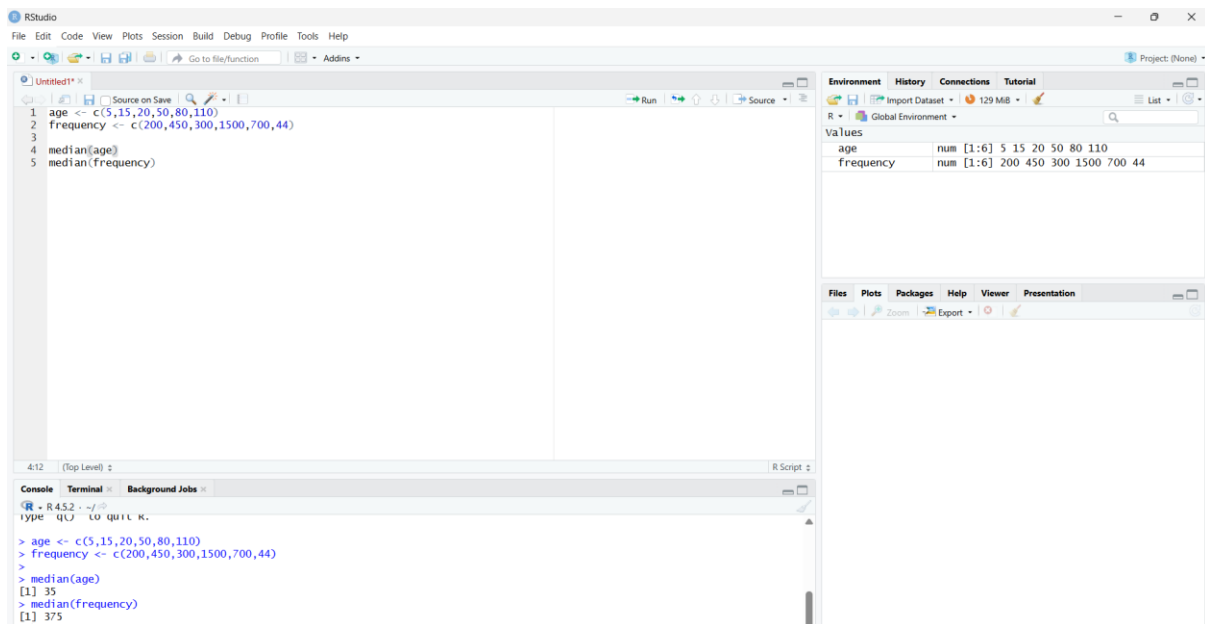
Q1. Approximate Median

Aim: To compute the approximate median value for grouped age and frequency data.

R-CODE:

```
age <- c(5,15,20,50,80,110)
frequency <- c(200,450,300,1500,700,44)
median(age)
median(frequency)
```

OUTPUT:



Q2. Mean, Median, Mode, Midrange & Quartiles

Aim: To calculate the mean, median, mode, midrange, first quartile, and third quartile of age data.

R-CODE:

```
age <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
mean(age)
median(age)
names(table(age))[table(age)==max(table(age))]
```

```
range(age)
```

```
quantile(age,0.25)
```

```
quantile(age,0.75)
```

OUTPUT:

```
10:1 (Top Level)
Console Terminal Background Jobs
R - R 4.5.2 - ~/
> age <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
>
> mean(age)
[1] 29.96296
> median(age)
[1] 25
> names(table(age))[table(age)==max(table(age))]
[1] "25" "35"
> range(age)
[1] 13 70
> quantile(age,0.25)
25%
20.5
> quantile(age,0.75)
75%
35
> |
```

Q3. Normalization

Aim: To perform Min–Max and Z-score normalization.

R-CODE:

```
x <- c(200,300,400,600,1000)
```

```
min_max <- (x-min(x))/(max(x)-min(x))
```

```
min_max
```

```
z_score <- (x-mean(x))/sd(x)
```

```
z_score
```

OUTPUT:

```
7:8 (Top Level)
Console Terminal Background Jobs
R - R 4.5.2 - ~/
> x <- c(200,300,400,600,1000)
>
> min_max <- (x-min(x))/(max(x)-min(x))
> min_max
[1] 0.000 0.125 0.250 0.500 1.000
>
> z_score <- (x-mean(x))/sd(x)
> z_score
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
> |
```

Q4. Smoothing by Binning

Aim: To perform smoothing using bin mean, bin median, and bin boundaries.

R-CODE:

```
data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,75)

bins <- 5

b <- cut(data,bins)

tapply(data,b,mean)

tapply(data,b,median)

tapply(data,b,function(x)c(min(x),max(x)))
```

OUTPUT:

```
> data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,75)
>
> bins <- 5
> b <- cut(data,bins)
>
> tapply(data,b,mean)
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
 17.78571 27.00000 43.75000 NA 72.75000
> tapply(data,b,median)
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
 19.5 27.0 45.0 NA 72.5
> tapply(data,b,function(x)c(min(x),max(x)))
$`(10.9,23.8]`
[1] 11 23

$`(23.8,36.6]`
[1] 24 30

$`(36.6,49.4]`
[1] 40 45

$`(49.4,62.2]`
NULL

$`(62.2,75.1]`
[1] 71 75
```

Q5. Statistical Plots

Aim: To calculate mean, median, standard deviation and draw boxplot, scatter plot, and Q-Q plot.

R-CODE:

```
age <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)

fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)

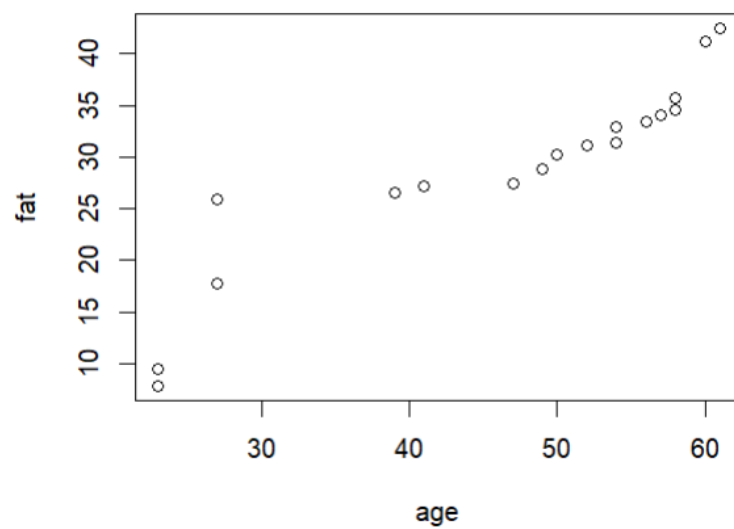
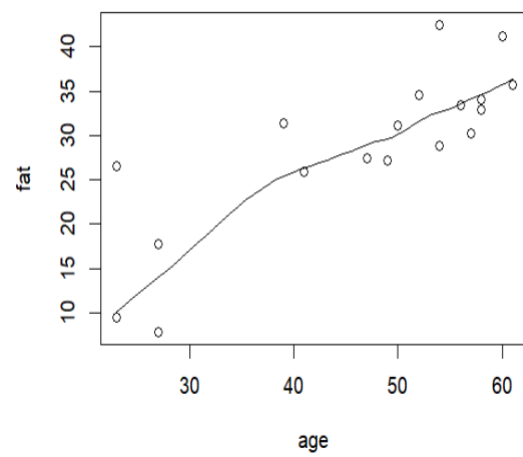
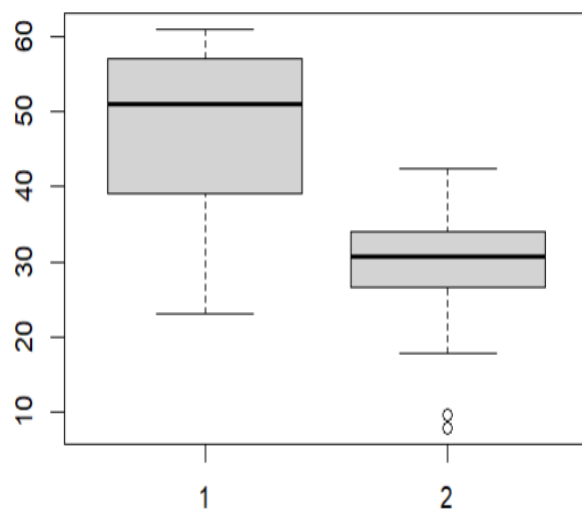
mean(age)

median(age)

sd(age)
```

```
mean(fat)
median(fat)
sd(fat)
boxplot(age,fat)
scatter.smooth(age,fat)
qqplot(age,fat)
```

OUTPUT:



```

R • R 4.5.2 • ~/
> age <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
> fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)
>
> mean(age)
[1] 46.44444
> median(age)
[1] 51
> sd(age)
[1] 13.21862
>
> mean(fat)
[1] 28.78333
> median(fat)
[1] 30.7
> sd(fat)
[1] 9.254395
>
> boxplot(age,fat)
> scatter.smooth(age,fat)
> qqplot(age,fat)
>

```

Q6. Data Normalization Methods

Aim: To perform Min–Max, Z-score, and Decimal Scaling normalization for age 35.

R-CODE:

```
v <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
```

```
min_max <- (35-min(v))/(max(v)-min(v))
```

```
min_max
```

```
m <- mean(v)
```

```
s <- 12.94
```

```
z_score <- (35-m)/s
```

```
z_score
```

```
decimal_scaling <- 35/100
```

```
decimal_scaling
```

OUTPUT:

```
12.10 (top Level)
Console Terminal x Background Jobs x
R 4.5.2 ~ /
> v <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
>
> min_max <- (35-min(v))/(max(v)-min(v))
> min_max
[1] 0.3157895
>
> m <- mean(v)
> s <- 12.94
> z_score <- (35-m)/s
> z_score
[1] -0.8844238
>
> decimal_scaling <- 35/100
> decimal_scaling
[1] 0.35
> |
```

Q7. Mean, Median & Mode of Pencils

Aim: To find the mean, median, and mode of pencil counts.

R-CODE:

```
pencils <- c(9,25,23,12,11,6,7,8,9,10)
mean(pencils)
median(pencils)
names(table(pencils))[table(pencils)==max(table(pencils))]
```

OUTPUT:

```
Console Terminal x Background Jobs x
R 4.5.2 ~ /
> pencils <- c(9,25,23,12,11,6,7,8,9,10)
>
> mean(pencils)
[1] 12
> median(pencils)
[1] 9.5
> names(table(pencils))[table(pencils)==max(table(pencils))]
[1] "9"
> |
```

Q8. Scatter Plot

Aim: To draw a scatter plot for mobile phones sold and money earned.

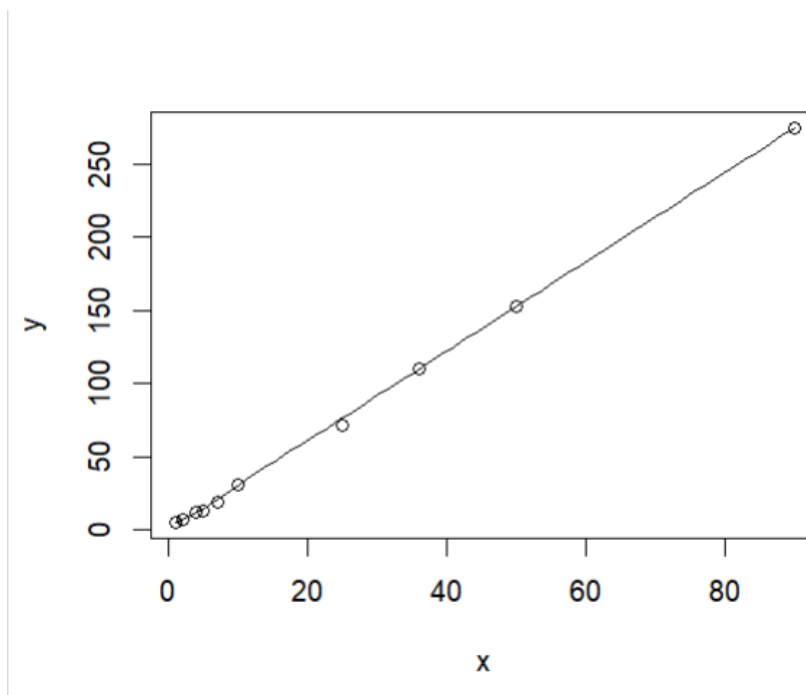
R-CODE:

```
x <- c(4,1,5,7,10,2,50,25,90,36)
```

```
y <- c(12,5,13,19,31,7,153,72,275,110)
```

```
scatter.smooth(x,y)
```

OUTPUT:



Q9. Partitioning & Histogram

Aim: To perform equal-frequency and equal-width partitioning and plot a histogram.

R-CODE:

```
marks <- c(55,60,71,63,55,65,50,55,58,59,61,63,65,67,71,72,75)
```

```
num_bins <- 3
```

```
cut(marks,breaks=num_bins,labels=FALSE)
```

```
hist(marks,breaks=num_bins,xlab="Marks",main="Histogram")
```

```
tapply(marks,cut(marks,num_bins),mean)
```

```
tapply(marks,cut(marks,num_bins),median)
```

```
tapply(marks,cut(marks,num_bins),function(x)c(min(x),max(x)))
```

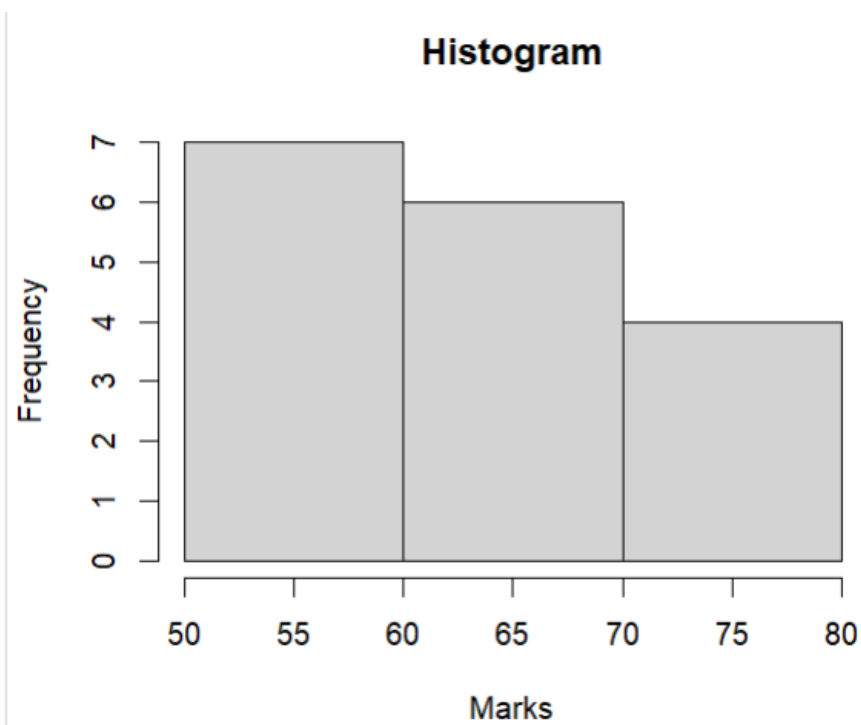
OUTPUT:

```
Console Terminal x Background Jobs x
R 4.5.2 · ~/
> cut(marks,breaks=num_bins,labels=FALSE)
[1] 1 2 3 2 1 2 1 1 1 2 2 2 2 3 3 3 3
>
> hist(marks,breaks=num_bins,xlab="Marks",main="Histogram")
>
> tapply(marks,cut(marks,num_bins),mean)
(50,58.3] (58.3,66.7] (66.7,75]
54.60000 62.28571 71.20000
> tapply(marks,cut(marks,num_bins),median)
(50,58.3] (58.3,66.7] (66.7,75]
55 63 71
> tapply(marks,cut(marks,num_bins),function(x)c(min(x),max(x)))
$`(50,58.3]`
[1] 50 58

$`(58.3,66.7]`
[1] 59 65

$`(66.7,75]`
[1] 67 75

>
```



Q10. Interquartile Range & Standard Deviation

Aim: To calculate the Interquartile Range (IQR) and Standard Deviation of car speeds.

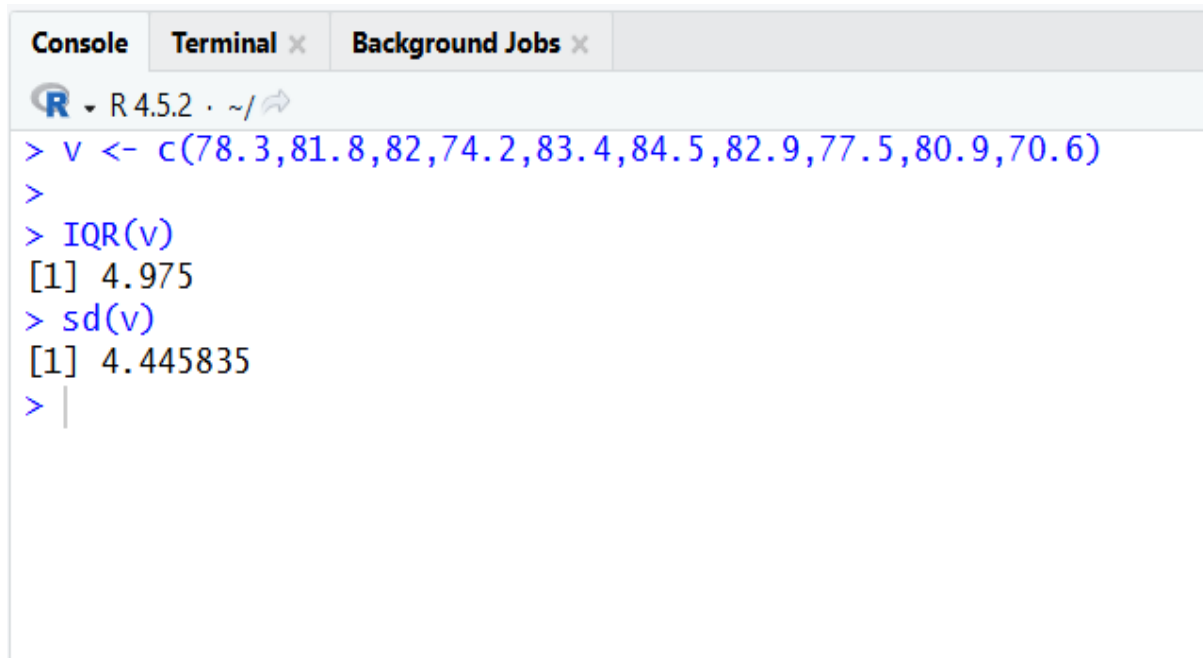
R-CODE:

```
v <- c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
```

```
IQR(v)
```

```
sd(v)
```

OUTPUT:

A screenshot of an R console window. The window has three tabs: 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active. The console shows the R prompt '>' followed by the command 'v <- c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)'. Below this, the prompt '>' is followed by 'IQR(v)', which returns '[1] 4.975'. Then, the prompt '>' is followed by 'sd(v)', which returns '[1] 4.445835'. The prompt '>' is followed by a vertical bar '|', indicating the command is still being processed or the output is being displayed.

```
Console Terminal x Background Jobs x
R 4.5.2 · ~/
> v <- c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
>
> IQR(v)
[1] 4.975
> sd(v)
[1] 4.445835
> |
```

Q11. Quartiles

Aim: To find the first quartile (Q1) and third quartile (Q3) of the age data.

R-CODE:

```
age <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
```

```
quantile(age,0.25)
```

```
quantile(age,0.75)
```

OUTPUT:

```
Console Terminal x Background Jobs x
R R 4.5.2 · ~/
> age <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
>
> quantile(age,0.25)
25%
20.5
> quantile(age,0.75)
75%
35
> |
```